

STRUCTURAL GENERAL NOTES

THESE NOTES SUPPLEMENT THE SPECIFICATIONS WHICH SHALL BE REFERRED TO FOR ADDITIONAL REQUIREMENTS. THESE NOTES APPLY TO CONTRACTORS, SUBCONTRACTORS, MANUFACTURERS, SUPPLIERS, FABRICATORS, ERECTORS, ETC. ENGAGED IN THE EXECUTION OF WORK INDICATED ON THESE DRAWINGS.

A. CODES AND STANDARDS

THE FOLLOWING CODES AND STANDARDS, INCLUDING ALL SPECIFICATIONS REFERENCED WITHIN, SHALL APPLY TO THE DESIGN, CONSTRUCTION, AND QUALITY CONTROL OF ALL WORK PERFORMED ON THE PROJECT. USE THE LATEST EDITION UNLESS NOTED OTHERWISE.

- BUILDING CODES:
 - "INTERNATIONAL BUILDING CODE", 2018 EDITION WITH 2020 AND 2022, 2024, AND 2025 GEORGIA STATE AMENDMENTS.
 - THE GWINNETT COUNTY CONSTRUCTION CODE, 2015 VERSION WITH 2024 AMENDMENTS.
 - "MINIMUM DESIGN LOADS FOR BUILDINGS AND OTHER STRUCTURES", ASCE 7-16.
- AMERICAN CONCRETE INSTITUTE (ACI):
 - "BUILDING CODE REQUIREMENTS FOR STRUCTURAL CONCRETE", ACI 318-14.
 - "ACI MANUAL OF CONCRETE PRACTICE - PARTS 1 THROUGH 5", 2020.
 - "BUILDING CODE REQUIREMENTS FOR MASONRY STRUCTURES" (ACI 530-16/ASCE 5-13/TMS 402-16) AND "SPECIFICATIONS FOR MASONRY STRUCTURES" (ACI 530.1-13/ASCE 6-13/TMS 602-16).
- CONCRETE REINFORCING STEEL INSTITUTE (CRSI)
 - "MANUAL OF STANDARD PRACTICE".
- AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)
 - "STEEL CONSTRUCTION MANUAL", FIFTEENTH EDITION, 2017.
 - "SPECIFICATION FOR STRUCTURAL STEEL BUILDINGS", AISC 360-16.
 - "CODE OF STANDARD PRACTICE FOR STEEL BUILDINGS AND BRIGES", AISC 303-16.
- RESEARCH COUNCIL ON STRUCTURAL CONNECTIONS (RCSC)
 - "SPECIFICATION FOR STRUCTURAL JOINTS USING ASTM A325 OR A490 BOLTS".
- AMERICAN WELDING SOCIETY (AWS)
 - "STRUCTURAL WELDING CODE - STEEL", ANSI/AWS D1.1-10.
 - "STRUCTURAL WELDING CODE - SHEET STEEL", ANSI/AWS D1.3-08.
- STEEL JOIST INSTITUTE (SJI)
 - "STANDARD SPECIFICATIONS FOR OPEN WEB STEEL JOISTS, K-SERIES".
 - "STANDARD SPECIFICATIONS FOR JOIST GIRDERS".
- STEEL DECK INSTITUTE
 - "STANDARD FOR STEEL ROOF DECK", ANSISDI RD-10.
- AMERICAN IRON AND STEEL INSTITUTE (AISI)
 - "NORTH AMERICAN SPECIFICATION FOR THE DESIGN OF COLD-FORMED STEEL STRUCTURAL MEMBERS", ANSIS100-16.

B. DESIGN DATA

- SUPERIMPOSED DEAD LOADS (IN ADDITION TO STRUCTURE SELF-WEIGHT)

AREA	PSE
a. ROOF	23
 - LIVE LOADS

AREA	PSE
a. ROOF LIVE LOAD SNOW LOAD:	20
1. GROUND SNOW LOAD (P_g):	5
2. SNOW EXPOSURE FACTOR (C_e):	1.0
3. SNOW LOAD IMPORTANCE FACTOR (I_s):	1.0
4. ROOF THERMAL FACTOR (C_t):	1.0
5. FLAT-ROOF SNOW LOAD: $P_f = 0.7 C_e C_p I_s$ (BUT NOT LESS THAN 20kps)	
 - WIND LOADS
 - RISK CATEGORY: II
 - WIND EXPOSURE CATEGORY: B
 - WIND SPEED:
 - ULTIMATE WIND SPEED, 3-SECOND GUST (V_{ult}): 107 MPH
 - NOMINAL WIND SPEED, 3-SECOND GUST (V_{nom}): 82.9 MPH
 - INTERNAL PRESSURE COEFFICIENT (GCP): +0.18; -0.18 (ENCLOSED BUILDING)

ANALYSIS PROCEDURE: ANALYTICAL PROCEDURE

BUILDING COMPONENTS AND CLADDING: SEE 2020 FOR DIAGRAMS OF ROOF AND WALL ZONES

 - MAXIMUM EXTERNAL PRESSURE COEFFICIENT (GCE) (BASED ON EFFECTIVE AREA = 10' R² AND LOCATED AT THE ROOF):

WALLS (ZONE 4):	+1.1; -1.2
WALLS (ZONE 5):	+1.1; -1.4
ROOFS (ZONE 1):	+0.3; -1.7
ROOFS (ZONE 2):	+0.3; -2.3
ROOFS (ZONE 3):	+0.3; -3.2
 - MAXIMUM ULTIMATE DESIGN PRESSURES (BASED ON EFFECTIVE AREA = 10' R² AND LOCATED AT THE ROOF):

WALLS (ZONE 4):	+16.0 PSF; -17.3 PSF
WALLS (ZONE 5):	+16.0 PSF; -21.3 PSF
ROOFS (ZONE 1):	+16.0 PSF; -27.9 PSF
ROOFS (ZONE 2):	+16.0 PSF; -36.8 PSF
ROOFS (ZONE 3):	+16.0 PSF; -50.1 PSF
 - COMPONENTS AND CLADDING LOADS VARY DUE TO THE ELEVATION AND EFFECTIVE AREA OF THE ELEMENT UNDER CONSIDERATION. REDUCED DESIGN PRESSURES ARE TO BE CALCULATED FOR ELEMENTS WITH LARGER EFFECTIVE AREAS OR LOCATED BELOW THE ROOF IN ACCORDANCE WITH ASCE 7-16.
- SEISMIC LOADS
 - RISK CATEGORY: II
 - SEISMIC IMPORTANCE FACTOR (I_s): 1.0
 - MAPPED SPECTRAL RESPONSE ACCELERATIONS:

$S_{DS} = 0.19$
$S_{D1} = 0.086$
 - SITE CLASSIFICATION: D
 - DESIGN SPECTRAL RESPONSE ACCELERATIONS:

$S_{DS} = 0.203$
$S_{D1} = 0.138$
 - SEISMIC DESIGN CATEGORY: C
 - SEISMIC FORCE RESISTING SYSTEM: INTERMEDIATE REINFORCED MASONRY SHEAR WALLS
 - RESPONSE MODIFICATION FACTOR (R): 3.5
 - ANALYSIS PROCEDURE: EQUIVALENT LATERAL FORCE PROCEDURE
 - SEISMIC RESPONSE COEFFICIENT (C_s): 0.058
 - DESIGN BASE SHEAR (V): = 20.7 KIPS

ARCHITECTURAL, MECHANICAL AND ELECTRICAL COMPONENTS AND SYSTEMS: DESIGN COMPLETE SYSTEM IN ACCORDANCE WITH CHAPTER 13 OF ASCE 7-16 AND ABOVE INFORMATION.
- LATERAL SOIL LOADS
 - LATERAL EQUIVALENT FLUID PRESSURE

1. AT REST CONDITION (BRACED WALLS)	60 PSF/FT OF DEPTH
2. ACTIVE CONDITION (CANTILEVERED RETAINING WALLS)	40 PSF/FT OF DEPTH
3. PASSIVE CONDITION	150 PSF/FT OF DEPTH
4. SLIDING RESISTANCE (FRICTION FACTOR)	0.35
 - NOTE THAT THE ABOVE NOTED LATERAL EQUIVALENT FLUID PRESSURES (PER THE RECOMMENDATIONS IN THE GEOLOGICAL REPORT) ASSUME 30' HYDROSTATIC PRESSURE ON THE WALLS, I.E. ADEQUATE SUBSURFACE DRAINAGE WILL BE PROVIDED.

C. FOUNDATIONS / GEOTECHNICAL REPORT

- FOUNDATIONS HAVE BEEN DESIGNED IN ACCORDANCE WITH THE RECOMMENDATIONS INCLUDED IN THE GEOTECHNICAL REPORT PREPARED BY NOVA ENGINEERING AND ENVIRONMENTAL, LLC (NOVA), DATED MARCH 26, 2021, REPORT NO. 10103-2021-008. REFER TO THAT REPORT FOR ADDITIONAL REQUIREMENTS.
- SLABS ON GRADE HAVE BEEN BASED ON A MODULUS OF SUBGRADE REACTION OF 125 POUNDS PER CUBIC INCH, BASED ON INFORMATION IN THE PROJECT GEOTECHNICAL REPORT.
- FOUNDATIONS PLACED ON UNDISTURBED SOIL AT ELEVATIONS INDICATED HAVE BEEN DESIGNED FOR A NET ALLOWABLE BEARING PRESSURE OF 2,000 PSF.

D. MATERIALS

THE FOLLOWING ASTM STANDARDS AND DESIGN STRESSES SHALL BE USED FOR THE APPROPRIATE MATERIALS USED IN THE CONSTRUCTION OF THIS PROJECT.

- CEMENT: ASTM C150, TYPE I, II, OR III
- BLENDED HYDRAULIC CEMENT: ASTM C596, TYPE IS (LIMIT SLAG TO 35% MAX CONTENT BY WEIGHT)
- AGGREGATES: ASTM C33 (NORMAL WEIGHT); 1" NOMINAL MAXIMUM AGGREGATE SIZE.
- ADMIXTURES: AIR ENTRAINING ADMIXTURES ASTM C260
CHEMICAL ADMIXTURES ASTM C494
- CONCRETE: AIR ENTRAIN CONCRETE AS INDICATED BELOW, AND CONCRETE EXPOSED TO EARTH AND OTHERS, 6% ± 1% BY VOLUME. UNLESS NOTED OTHERWISE, "EXTENDED" MEMBERS ARE THOSE FULLY OR PARTIALLY OUTSIDE OF THE CONDITIONED BUILDING ENVELOPE AND FULLY OR PARTIALLY ABOVE THE FROST DEPTH.

APPLICATION	f_c @ 28 DAYS (PSI)	WT (PCF)	W/C RATIO	AIR (%)	ENTRAIN
a. FOOTINGS	3000	145	F0	0.55	Y
b. EXTERIOR WALLS & PIERS	4500	145	F2	0.45	Y
c. INTERIOR SLABS-ON-GRADE	4000	145	F0	0.55	N
- REINFORCEMENT:
 - DEFORMED REINFORCING BARS ASTM A615, GRADE 60
- MASONRY:
 - MASONRY COMPRESSIVE STRENGTH (F_m) 2000 PSI (UNIT STRENGTH METHOD)
 - LOAD BEARING CONCRETE MASONRY UNITS HOLLOW - ASTM C90; FINISH AND COLOR PER SPECIFICATIONS; MINIMUM COMPRESSIVE STRENGTH ON NET AREA = 2000 PSI, NORMAL WEIGHT (135 PCF)
 - MORTAR ASTM C270 - TYPE S
 - GROUT ASTM C476; MINIMUM COMPRESSIVE STRENGTH AT 28 DAYS = 2000 PSI; SLUMP BETWEEN 8" AND 11"
 - HORIZONTAL JOINT REINFORCING ASTM A951 $F_y=70$ KSI, 9 GAGE TRUSS-TYPE GALVANIZED
- STEEL:
 - WIDE FLANGE OTHER STRUCTURAL SHAPES AND PLATES ASTM A36
 - HOLLOW STRUCTURAL SECTIONS ASTM A500, GRADE B, $F_y=46$ KSI
 - COLD-FORMED STEEL FRAMING ASTM A653 (G-60 GALVANIZED)
 - MATERIAL 43 MILS AND THINNER $F_y = 33$ KSI (MIN.)
 - MATERIAL 54 MILS AND THICKER $F_y = 50$ KSI (MIN.)
 - GC NOTE: WHEN ORDERING MATERIAL, STEEL STRENGTHS LISTED MUST BE EXPLICITLY REQUESTED AS STOCK MATERIALS MAY BE AVAILABLE IN MULTIPLE GRADES.
 - HIGH-STRENGTH BOLTS ASTM F3125
 - ANCHOR RODS ASTM F1554, GRADE 36
 - SMOOTH AND THREADED ROD ASTM A36
 - HEADED SHEAR STUDS ASTM A108
 - WELDING ELECTRODES AWS A5.1 OR A5.5, E70XX (STRUCTURAL STEEL), E60XX (COLD-FORMED STEEL AND STEEL DECK)
 - GALVANIZED STEEL ROOF DECK GROUT ASTM A653, G-90 GALVANIZED, $F_y = 33$ KSI MIN.
 - HOT-DIPPED GALVANIZING ASTM A123 / ASTM A153 / ASTM F2329

E. CONSTRUCTION

- GENERAL:
 - THESE DRAWINGS REPRESENT THE COMPLETED PROJECT WHICH HAS BEEN DESIGNED FOR THE LOADS INDICATED IN THE DESIGN DATA. IT IS THE CONTRACTOR'S RESPONSIBILITY TO DETERMINE PROPER ERECTION PROCEDURES, ALLOWABLE CONSTRUCTION SEQUENCES, AND PROPER DESIGN AND CONSTRUCTION OF FORMWORK, BRACING, SHEETING AND SHORING, ETC.
 - IMPLEMENTING JOB SITE SAFETY AND CONSTRUCTION PROCEDURES ARE SOLELY THE RESPONSIBILITY OF THE CONTRACTOR.
 - THE STRUCTURAL DRAWINGS ARE INTENDED TO INDICATE THE MAIN STRUCTURAL FEATURES FOR THE PROJECT. ANY NON-STRUCTURAL (ARCHITECTURAL, MECHANICAL, ELECTRICAL, ETC.) DETAILS SHOWN ARE SCHEMATIC IN NATURE AND MAY NOT REFLECT THE COMPLETE CONSTRUCTION. THE PROJECT ARCHITECTURAL, CIVIL, AND MEP DRAWINGS MUST BE USED IN CONJUNCTION WITH THE PROJECT STRUCTURAL DRAWINGS DURING ALL PHASES OF CONSTRUCTION. ANY DISCREPANCIES BETWEEN THE PROJECT STRUCTURAL DRAWINGS AND THE SPECIFICATIONS AND DRAWINGS OF THE OTHER DISCIPLINES SHALL BE BROUGHT TO THE ATTENTION OF THE ARCHITECT PRIOR TO COMMENCING THE WORK.
 - REFER TO ARCHITECTURAL, MECHANICAL, PLUMBING, AND ELECTRICAL DRAWINGS FOR SIZE AND LOCATIONS OF OPENINGS WITH THE CONTRACT DOCUMENTS AND PADS, INSERTS, EMBEDS, CURBS, RAMPS, DRAINS, DEPRESSIONS, ETC.
 - REFER TO ARCHITECTURAL DRAWINGS FOR DIMENSIONS NOT SHOWN.
 - REFER TO ARCHITECTURAL DRAWINGS AND SPECIFICATIONS FOR DETAILED INFORMATION REGARDING FINISHES, FIRE RATINGS, FIREPROOFING, WATERPROOFING, ETC.
 - DO NOT SCALE THE STRUCTURAL DRAWINGS.
 - REFER TO ARCHITECTURAL DRAWINGS FOR LOCATIONS OF MASONRY AND DRYWALL NON-LOAD BEARING PARTITIONS. PROVIDE SLIP CONNECTIONS THAT ALLOW VERTICAL MOVEMENT OF THE HEADS OF SUCH WALLS. DESIGN CONNECTIONS TO SUPPORT THE TOPS OF THE WALLS LATERALLY FOR THE CODE-REQUIRED LATERAL LOAD.
 - IN CASE OF CONFLICT BETWEEN THE GENERAL NOTES, SPECIFICATIONS, AND DRAWINGS, THE MOST RIGID REQUIREMENTS GOVERN.
 - STORE AND HANDLE STRUCTURAL CONSTRUCTION MATERIALS TO PREVENT ANY ADVERSE EFFECTS ON THE PHYSICAL PROPERTIES OF THE MATERIAL.
 - PAY ALL COSTS, INCLUDING INVESTIGATION AND/OR REDESIGN, DUE TO CONTRACTOR MISLOCATION OF STRUCTURAL ELEMENTS OR OTHER LACK OF CONFORMANCE WITH THE CONTRACT DOCUMENTS TO BRING WORK IN COMPLIANCE WITH THE CONTRACT DOCUMENTS.
- SHOP DRAWINGS AND SUBMITTALS:
 - REPRODUCTION OF ANY PORTION OF THE STRUCTURAL CONSTRUCTION DRAWINGS FOR RESUBMITTAL AS SHOP DRAWINGS IS PROHIBITED. SHOP DRAWINGS PRODUCED IN SUCH A MANNER WILL BE REJECTED WITHOUT REVIEW AND RETURNED.
 - SUBMIT SHOP DRAWINGS AT LEAST 15 BUSINESS DAYS BEFORE DATE REVIEWED SUBMITTALS WILL BE NEEDED. SHOP DRAWINGS SHALL BEAR THE CONTRACTOR'S STAMP OF APPROVAL WHICH SHALL CONSTITUTE CERTIFICATION THAT THE CONTRACTOR HAS VERIFIED ALL FIELD MEASUREMENTS, CONSTRUCTION CRITERIA, MATERIALS AND SIMILAR DATA AND HAS CHECKED EACH DRAWING FOR COMPLETENESS, COORDINATION AND COMPLIANCE WITH THE CONTRACT DOCUMENTS.
- STRUCTURAL COLD-FORMED STEEL FRAMING
 - MEMBER DESIGNATIONS AND PROPERTIES ARE BASED ON STEEL STUD MANUFACTURERS ASSOCIATION STANDARDS. FRAMING MEMBERS SUPPLIED SHALL HAVE SECTION PROPERTIES WHICH EQUAL OR EXCEED THE DESIGNATED SECTION PROPERTIES.
 - ALL COLD-FORMED STEEL FRAMING AND ACCESSORIES SHALL BE G60 GALVANIZED AFTER ROLLING TO THE REQUIRED THICKNESS. ALL STEEL SHALL BE MILL CERTIFIED FOR COMPLIANCE WITH STRENGTH AND COATING SPECIFICATIONS.
 - CONNECTIONS SHALL BE BY SCREWS, BOLTS OR OTHER APPROVED FASTENING DEVICES OR METHODS. PROVIDE POSITIVE ATTACHMENT AND RESISTANCE TO LOOSENING. FASTENERS SHALL BE OF COMPATIBLE MATERIAL AND COATED FOR CORROSION RESISTANCE AGAINST EXTERIOR EXPOSURES.
 - FIELD CUTTING OF CFS MEMBERS SHALL BE BY SAW OR SHEAR. FLAME CUTTING IS NOT PERMITTED.
 - BRACE CFS INSTALLATIONS AS NEEDED UNTIL THE WORK IS STABILIZED.
 - PAINT ALL DAMAGED AREAS OF GALVANIZED COATING, AND CUT ENDS WITH GALVANIZING REPAIR PAINT COMPLYING WITH FEDERAL SPECIFICATION MIL-P-21035A (REPAIR OF HOT-DIP GALVANIZING).
 - FASTEN INDIVIDUAL ELEMENTS OF BUILT-UP SECTIONS TOGETHER AT NO MORE THAN 16" O.C. UNLESS NOTED OTHERWISE.
 - PROVIDE BRIDGING FOR ALL MEMBERS AT 5'-0" O.C. MAXIMUM UNLESS NOTED OTHERWISE. MEMBERS LESS THAN 5'-0" LONG DO NOT REQUIRE BRIDGING.
 - FOR SCREW CONNECTIONS, PROVIDE A MINIMUM OF (2) #10 SCREWS FOR EACH CONNECTION UNLESS NOTED OTHERWISE.
 - SELECT SCREWS WITH ADEQUATE CUTTING TIP TO PENETRATE TOTAL THICKNESS OF MATERIAL. TO BE DRILLED. SCREWS SHALL PENETRATE THE THINNER MEMBER BEING FASTENED FIRST AND SHALL HAVE A MINIMUM OF THREE THREADS EXPOSED ON THE FAR SIDE OF THE CONNECTION.
 - SCREWS SHALL MAINTAIN A MINIMUM EDGE DISTANCE OF 3/4" AND A MINIMUM SPACING OF 1".

SUBMIT CALCULATIONS AND DRAWINGS CONCURRENTLY FOR EACH OF THE FOLLOWING ASSEMBLIES. DESIGN EACH ASSEMBLY UNDER THE DIRECT SUPERVISION OF AN ENGINEER REGISTERED IN THE PROJECT'S JURISDICTION. ALL PORTIONS OF SUBMITTALS SHALL BEAR THIS ENGINEER'S SEAL AND SIGNATURE. UNSEALED SUBMITTALS WILL BE RETURNED AND REJECTED WITHOUT REVIEW. REVIEW SHALL BE FOR GENERAL CONFORMANCE WITH THE PROJECT PARAMETERS AS INDICATED ON THE DRAWINGS AND IN THE GENERAL NOTES AND SPECIFICATIONS, AND FOR IMPACTS ON THE SUPPORTING STRUCTURAL SYSTEM. DESIGN FOR ALL GRAVITY AND LATERAL LOADS AND OTHER EFFECTS INCLUDING CREEP, SHRINKAGE, THERMAL, ETC.) REQUIRED BY APPLICABLE CODES AND STANDARDS AS WELL AS THOSE INDICATED ON THE DRAWINGS.

- NON-LOAD BEARING STUD WALL AND RELATED CONNECTIONS TO THE STRUCTURE
 - UNLESS SPECIFICALLY NOTED OTHERWISE, PROVIDE SLIP CONNECTIONS THAT ALLOW VERTICAL MOVEMENT AT THE HEADS OF WALLS.
 - INTERIOR PARTITION WALLS: DESIGN SYSTEM FOR A MAXIMUM HORIZONTAL DEFLECTION OF 1/280 OF THE SPAN IN INCHES, OR 0.50", WHICHEVER IS LESS, AT THE APPLICABLE DESIGN LATERAL LOAD.
 - CURTAINWALL/WINDOW-WALL SYSTEMS AND RELATED CONNECTIONS TO THE STRUCTURE
 - PROVIDE SLIP CONNECTIONS THAT ALLOW VERTICAL MOVEMENT AT THE HEADS OF WALLS.
 - DESIGN CURTAINWALL FOR A MAXIMUM HORIZONTAL DEFLECTION THAT IS COMPATIBLE WITH THE WINDOW/METAL PANEL SYSTEM DEFLECTION LIMITATIONS (1/280 OF THE SPAN IN INCHES, OR 0.50", WHICHEVER IS LESS, AT THE APPLICABLE DESIGN WIND LOAD.
 - STEEL JOISTS AND JOIST GIRDERS
 - DESIGN ALL JOISTS FOR THE LOADINGS AND CRITERIA INDICATED ON THE DRAWINGS. THESE NOTES, AND THE SPECIFICATIONS.
 - SUBMIT SIGNED AND SEALED CALCULATIONS FOR ALL SPECIAL (SP) JOISTS CONCURRENTLY WITH THE SHOP DRAWINGS.
 - DESIGN AND DETAIL JOIST BRIDGING AND CONNECTIONS AS REQUIRED FOR ERECTION STABILITY, NET UPLIFT, AND PER THE SJI'S SPECIFICATIONS.
 - CANOPY
 - DESIGN CANOPY FOR THE LOADINGS AND CRITERIA INDICATED ON THE DRAWINGS, THESE NOTES, AND THE SPECIFICATIONS.
 - SUBMITTALS AND CERTIFICATIONS. IN ADDITION TO STANDARD INDUSTRY PRACTICE
 - CAST-IN-PLACE CONCRETE
 - SHOP DRAWINGS INDICATING PROPOSED JOINT LOCATIONS AND REINFORCING STEEL TO BE PLACED IN THE SLAB ON GRADE, CONCRETE MIX DESIGNS, INCLUDING DOCUMENTATION USED TO DETERMINE STANDARD DEVIATION IN ACCORDANCE WITH ACI 301. IF CEMENTITIOUS SUBSTITUTES OR ADMIXTURES ARE TO BE USED, PROVIDE DOCUMENTATION IN THE MIX DESIGN THAT INDICATES CONFORMANCE TO THE MAXIMUM PERCENTAGE OF SUBSTITUTE. CERTIFICATIONS OF REINFORCING STEEL COMPLIANCE WITH REFERENCED STANDARDS.
 - MASONRY
 - MASONRY GROUT MIX DESIGN AND MASONRY UNIT CERTIFICATIONS.
 - REINFORCING STEEL SHOP DRAWINGS IN REINFORCED MASONRY CONSTRUCTION. INDICATE PROPOSED CONTROL JOINT LOCATIONS.
 - CERTIFICATIONS OF REINFORCING STEEL COMPLIANCE WITH REFERENCED STANDARDS.
 - STRUCTURAL STEEL
 - CERTIFIED COPIES OF MILL TEST REPORTS FOR RECORD PURPOSES ONLY.
 - ALL WELD PROCEDURE SPECIFICATIONS THAT ARE QUALIFIED BY TEST (IN ACCORDANCE WITH AWS D1.1), ACCOMPANIED BY PROCEDURE QUALIFICATION RECORDS AND ELECTRODE DATA SHEETS.
 - INSPECTION AND TESTING:
 - THE OWNER WILL ENGAGE AN APPROVED, QUALIFIED, TESTING AGENCY TO PROVIDE TESTING AND INSPECTION SERVICES. IN ADDITION TO THE SPECIAL INSPECTION REQUIREMENTS AS INDICATED BELOW, SUBMIT REPORTS TO THE SER AND CODE OFFICIALS (AS APPLICABLE).
 - REQUIRED INSPECTIONS ARE SHOWN ON DRAWING
 - CONTRACTOR SHALL REGULARLY PROVIDE SER WITH A CURRENT CONSTRUCTION SCHEDULE SO THAT THE REQUIRED INSPECTIONS CAN BE PROVIDED IN A TIMELY MANNER
 - SOILS TESTING
 - TEST ALL DIFFERING FOUNDATION SUBGRADE SOIL STRATA OBSERVED IN A MINIMUM OF ONE LOCATION FOR EACH FOR MINIMUM REQUIRED BEARING CAPACITY (INDICATED EARLIER IN THESE NOTES). SUBSEQUENT VERIFICATION OF BEARING CAPACITY AT OTHER LOCATIONS MAY BE BY VISUAL COMPARISON WITH TESTED LOCATIONS.
 - TESTING AGENCY WILL TEST COMPACTION OF STRUCTURAL FILL PER ASTM D1556, ASTM D2167, ASTM D2922, AND ASTM D2937, AS APPLICABLE. TEST A MINIMUM OF THREE LOCATIONS RANDOMLY LOCATED ACROSS THE BUILDING PAD. TEST ADDITIONAL AREAS IF POOR CONDITIONS ARE OBSERVED.
 - NOTIFY CONTRACTOR AND A/E OF AREAS FAILING TO MEET DESIGN REQUIREMENTS AND WHICH REQUIRE RECOMPARATIONS AND REINSPECTION.
 - CAST-IN-PLACE CONCRETE:
 - THE AGENCY SHALL INSPECT THE FORMWORK AND REINFORCING STEEL PLACEMENT FOR COMPLIANCE WITH THE CONTRACT DOCUMENTS AND SHOP DRAWINGS. THE AGENCY SHALL MONITOR ALL STRUCTURAL CONCRETE PLACEMENTS FOR CONFORMANCE WITH APPLICABLE ACI REQUIREMENTS.
 - SAMPLE FRESH CONCRETE IN ACCORDANCE WITH ASTM C172. MOLD TEST CYLINDERS IN ACCORDANCE WITH ASTM C31.
 - UNLESS OTHERWISE NOTED AND/OR APPROVED IN WRITING BY THE SER, SAMPLING FOR TESTING AND INSPECTION OF CONCRETE SHALL BE AT THE POINT OF PLACEMENT.
 - THE FOLLOWING NUMBER OF 6"x12" ** TEST CYLINDERS SHALL BE CAST FOR EACH DAY'S POUR OR EACH 50 CUBIC YARDS, WHICHEVER RESULTS IN MORE TEST CYLINDERS:

2 @ 7 DAYS, LAB CURED
2 @ 7 DAYS, FIELD CURED
2 @ 28 DAYS, LAB CURED
2 @ 28 DAYS, FIELD CURED
2 @ 56 DAYS, LAB CURED*

* 28-DAY BREAKS ARE THE STANDARD FOR CONCRETE ACCEPTANCE IN THE FINAL STRUCTURE. RESERVE 56-DAY CYLINDERS FOR ADDITIONAL TESTING AT LOW 28-DAY BREAKS.

** 4"x6" CYLINDERS MAY BE USED FOR ACCEPTANCE TESTING IN LIEU OF 6"x12" CYLINDERS WITH THE PRIOR WRITTEN APPROVAL OF THE SER. WHERE SUCH APPROVAL IS OBTAINED, A MINIMUM OF THREE (3) 4"x6" CYLINDERS IS REQUIRED TO BE TAKEN AND TESTED AS A REPLACEMENT FOR TWO (2) 6"x12" CYLINDERS.
 - THE AGENCY SHALL OBTAIN AND TEST FIELD-CORED SAMPLES OF IN-PLACE CONCRETE AT THE CONTRACTOR'S EXPENSE WHEN TEST RESULTS INDICATED SPECIFIED CONCRETE STRENGTHS HAVE NOT BEEN ATTAINED. CORING LOCATIONS AND QUANTITIES SHALL BE DIRECTED BY THE SER.
- MASONRY:
 - THE AGENCY SHALL MONITOR THE PROPORTIONING, MIXING AND CONSISTENCY OF MORTAR AND GROUT, INSTALLATION OF MASONRY UNITS, SIZE AND LOCATION OF REINFORCEMENT, ANCHORAGE OF MASONRY, AND PLACEMENT OF MORTAR AND GROUT FOR COMPLIANCE WITH THE CONTRACT DOCUMENTS.
 - COMPRESSION TEST MASONRY PRISMS FOR EACH TYPE OF WALL CONSTRUCTION IN ACCORDANCE WITH ASTM E447, METHOD B.
 - THE CONTRACTOR SHALL PREPARE ONE SET OF PRISMS FOR TESTING AT 7 DAYS AND ONE SET FOR TESTING AT 28 DAYS. TESTS ARE TO BE CONDUCTED BY THE AGENCY FOR EACH 5000 SQUARE FEET OF WALL INSTALLED, BUT NOT LESS THAN 2 TESTS.
- STRUCTURAL STEEL:
 - THE AGENCY SHALL REVIEW PREQUALIFIED WELD PROCEDURE SPECIFICATIONS IN ACCORDANCE WITH AWS D1.1, SECTION 6.3.1.
 - THE AGENCY SHALL VISUALLY INSPECT ALL FILLET WELDS, AND BOLTED CONNECTIONS.
 - THE AGENCY SHALL PERFORM WELDING INSPECTION AND TESTING PROCEDURES IN ACCORDANCE WITH THE AWS CODE.
 - TEST EACH FULL PENETRATION BUTT OR GROOVE WELD AND 50% OF PARTIAL PENETRATION WELDS BY THE ULTRASONIC METHOD ASTM E164.
 - TEST 10% OF ALL FIELD FILLET WELDS IN PRIMARY CONNECTIONS AND MULTI-PASS WELDS BY THE MAGNETIC PARTICLE METHOD ASTM E709.
 - TEST ANY WELD FOR WHICH VISUAL EXAMINATION INDICATES AN UNUSUAL CONDITION AND/OR POOR QUALITY.
- POST-INSTALLED ANCHORS
 - ALL POST-INSTALLED FASTENERS SHALL BE VISUALLY INSPECTED FOR COMPLIANCE WITH ANCHOR TYPE, EMBEDMENT, SPACING, EDGE DISTANCE, AND ALL OTHER REQUIREMENTS AS INDICATED IN THE DRAWINGS AND ICC-ES REPORTS.
 - INSPECTOR TO CONFIRM THAT THE INSTALLATION OF ADHESIVE ANCHORS HORIZONTALLY OR UPWARDLY INCLINED IS PERFORMED BY PERSONNEL CERTIFIED BY AN APPLICABLE CERTIFICATION PROGRAM. CERTIFICATION SHALL INCLUDE WRITTEN AND PERFORMANCE TESTS IN ACCORDANCE WITH THE ACI/ACI'S ADHESIVE ANCHOR INSTALLER CERTIFICATION PROGRAM, OR EQUIVALENT.

F. FOUNDATIONS AND STRUCTURAL EARTHWORK

- GENERAL:
 - REFER TO PROJECT SPECIFICATION AND GEOTECHNICAL REPORT REQUIREMENTS FOR EXCAVATION AND PREPARATION OF THE FOUNDATION AND SLAB-ON-GRADE SUBGRADES, INCLUDING COMPACTION PROCEDURES. REQUIREMENTS CONTAINED IN THE GEOTECHNICAL REPORT ARE PART OF THIS WORK.
 - VERIFY ALL EXISTING FIELD CONDITIONS THAT MAY AFFECT THE INSTALLATION OF THE FOUNDATION SYSTEM AS SHOWN PRIOR TO STARTING WORK.
 - LOCATE AND PROTECT ALL EXISTING UTILITIES, WHETHER INDICATED OR NOT, WHICH MAY BE AFFECTED BY THE CONSTRUCTION PROCESS.
 - PROTECT EXISTING AND NEW STRUCTURES FROM DAMAGE BY CONSTRUCTION EQUIPMENT. REPAIR DAMAGE OF EXISTING AND NEW CONSTRUCTION CAUSED BY CONSTRUCTION TECHNIQUES.
 - DO NOT PLACE UTILITY LINES THROUGH OR BELOW FOUNDATIONS WITHOUT THE APPROVAL OF THE SER.
 - BEAR ALL FOUNDATIONS ON UNDISTURBED SOIL OR COMPACTED STRUCTURAL FILL. DETERMINATION OF FINAL BEARING ELEVATIONS AND FIELD VERIFICATION OF ALLOWABLE BEARING PRESSURE SHALL BE MADE BY A LICENSED GEOTECHNICAL ENGINEER. REPAIR DAMAGE OF EXISTING AND NEW CONSTRUCTION CAUSED BY CONSTRUCTION TECHNIQUES.
 - BEAR FOUNDATIONS A MINIMUM OF 2'-6" BELOW GRADE UNLESS OTHERWISE INDICATED. IN CASE OF CONFLICT, NOTIFY THE ARCHITECT AND SER IN ADVANCE OF ANY CONSTRUCTION TO ALLOW FOR ADJUSTMENT.
 - THE SLOPE BETWEEN THE LOWER EDGES OF ADJACENT FOOTINGS NOT TO EXCEED 1:5H:1.0V, UNLESS INDICATED OTHERWISE IN THE GEOTECHNICAL REPORT. PROTECT SUBGRADES, SLOPES AND FOOTINGS FROM DAMAGE CAUSED BY LATERAL MOVEMENT, UNDERMINING, SETTLEMENT AND OTHER HAZARDS CREATED BY EXCAVATION.
 - DO NOT USE EARTH CUTS AS FORMS FOR VERTICAL SURFACES UNLESS APPROVED IN ADVANCE BY THE SER.
 - PLACE CONCRETE FOR FOUNDATIONS OR MUD SLABS ON THE SAME DAY SUBGRADE APPROVAL IS GIVEN BY THE GEOTECHNICAL ENGINEER.
 - PROTECT CONCRETE FOUNDATIONS FROM FREEZING DURING PLACING AND FOR A PERIOD OF NOT LESS THAN 5 DAYS THEREAFTER.
- BACKFILL:
 - USE FREE-DRAINING BACKFILL MATERIAL APPROVED BY THE GEOTECHNICAL ENGINEER, WITH OPTIMUM MOISTURE CONTENT FOR COMPACTING AND FREE OF ANY DEBRIS.
 - WHERE THE FINAL GRADE ELEVATIONS ARE APPROXIMATELY EQUAL ON BOTH SIDES OF A WALL, BACKFILL IN LIFTS TO MAINTAIN LEVEL ELEVATIONS WITHIN 12" ON BOTH SIDES AT ANY TIME TO PREVENT LATERAL MOVEMENT AND/OR OVERTURNING.

G. CAST-IN-PLACE CONCRETE

- GENERAL:
 - COMPLY WITH REQUIREMENTS OF "STANDARD SPECIFICATIONS FOR STRUCTURAL CONCRETE" (ACI 301), EXCEPT AS MODIFIED BY THESE NOTES AND PROJECT SPECIFICATIONS. KEEP COPY OF "ACI FIELD REFERENCE MANUAL, SP-15" IN FIELD OFFICE.
 - PROVIDE MINIMUM CLEAR COVER FOR REINFORCING AS FOLLOWS, UNLESS OTHERWISE NOTED IN THE DRAWINGS:
 - CONCRETE CAST AGAINST AND PERMANENTLY IN CONTACT WITH GROUND: ALL BARS - 3"
 - CONCRETE EXPOSED TO WEATHER OR IN CONTACT WITH GROUND: #6 BARS AND LARGER - 2" #5 BARS AND SMALLER - 1 1/2"
 - SPlice REINFORCEMENT AS DETAILED OR AUTHORIZED BY THE SER. MAKE BARS CONTINUOUS AROUND CORNERS. SPLICES SHALL BE MADE BY CONTACT TENSION LAP SPLICES, UNLESS OTHERWISE NOTED.
 - WELDING OF REINFORCING IS NOT PERMITTED.
 - FIELD BENDING OF REINFORCING PARTIALLY EMBEDDED IN CONCRETE IS NOT PERMITTED UNLESS OTHERWISE SHOWN OR APPROVED BY THE SER.
 - FURNISH ALL ACCESSORIES, CHAIRS, SPACE BARS, SUPPORTS, ETC. NECESSARY TO SECURE REINFORCING.
 - CAST ALL INSERTS AND SLEEVES IN-PLACE.
 - PLACING SLEEVES THROUGH CONCRETE ELEMENTS IS ONLY PERMITTED WHEN SPECIFIED ON THE STRUCTURAL DRAWINGS. APPROVED SLEEVING SHOP DRAWINGS OR SPECIFICALLY AUTHORIZED IN WRITING BY THE SER.
 - ANY STOP IN CONCRETE WORK MUST BE MADE WITH VERTICAL BULKHEADS AND KEYS, UNLESS OTHERWISE SHOWN.
 - HORIZONTAL JOINTS ARE NOT PERMITTED IN FOUNDATIONS AND SLABS.
 - FINISH CONCRETE SLABS FLAT AND LEVEL WITHIN TOLERANCE, TO THE ELEVATION INDICATED ON THE DRAWINGS.
 - CORE DRILLING OF ANY CONCRETE ELEMENT IS NOT PERMITTED UNLESS AUTHORIZED IN WRITING BY THE SER.
 - CHAMFER EXPOSED CONCRETE CORNERS, 1/2" x 1/2" MINIMUM, UNLESS OTHERWISE NOTED ON THE DRAWINGS.
 - IN SLABS ON GRADE, PROVIDE 2#x5'-0" AT REENTRANT CORNERS UNLESS SPECIFICALLY NOTED OTHERWISE.
 - WHERE CONSTRUCTION JOINT IS NOTED AS "ROUGHENED" OR "ROUGHENED SURFACE", THE PREVIOUSLY PLACED CONCRETE SURFACE SHALL BE ROUGHENED TO A FULL AMPLITUDE OF APPROXIMATELY 1/2" AND BE FREE OF ALL LOOSE MATERIAL AND LAFTANCE.
 - WHEN INSTALLING POST-INSTALLED ANCHORS (EXPANSION BOLTS, ADHESIVE ANCHORS, ETC.) TAKE MEASURES TO AVOID DAMAGE TO EXISTING REINFORCING STEEL AND CONCRETE. CLEAN HOLES IN ACCORDANCE WITH MANUFACTURER'S REQUIREMENTS.

H. MASONRY

- WALL SECTIONS AND PIERS LESS THAN 2.0 SQUARE FEET IN CROSS-SECTIONAL AREA TO BE FULLY GROUTED WITH VERTICAL #4 REINFORCING BAR CENTERED IN EACH GROUTED CELL.
- IN GROUTED AND/OR REINFORCED MASONRY WALLS, USE MASONRY UNITS WITH CORES THAT ALIGN VERTICALLY TO PROVIDE CONTINUOUS UNOBSTRUCTED CELLS FOR GROUTING AND REINFORCING STEEL PLACEMENT.
- LAP SPLICES FOR DEFORMED REINFORCING BARS USED IN MASONRY CONSTRUCTION TO BE AS FOLLOWS:

43" 20"
43" 20"
43" 20"
43" 20"
- "WET-STICKING" OF REINFORCING STEEL IS NOT PERMITTED. USE PREFABRICATED METAL REBAR POSITONERS TO SECURE REINFORCING STEEL PRIOR TO GROUTING.
- COORDINATE GROUT PLACEMENT AND POUR/LIFT HEIGHTS IN ACCORDANCE WITH ACI 530.1 SECTION 3.5.
- CONSOLIDATE AND RECONSOLIDATE GROUT POURS EXCEEDING 12" IN HEIGHT BY MECHANICAL VIBRATION IN ACCORDANCE WITH ACI 530.1, SECTION 3.5.E.
- IN CAVITY WALLS, BOND THE WYTHES TOGETHER WITH RIGID METAL TIES OR PREFABRICATED JOINT REINFORCEMENT CONFORMING TO ACI 530 REQUIREMENTS. COMPLETELY FILL ALL COLLAR JOINTS IN ALL COMPOSITE WALLS AND ALL COLLAR JOINTS BELOW GRADE IN CAVITY WALLS WITH MORTAR OR GROUT.
- PROVIDE 8-GAGE GALVANIZED HORIZONTAL JOINT REINFORCEMENT IN ALL WALLS AND PARTITIONS AT 16" O.C. UNLESS OTHERWISE SHOWN OR NOTED. PROVIDE ONE PIECE PREFABRICATED UNITS AT 8" O.C. AT ALL WALL CORNERS AND INTERSECTIONS.
- ANCHOR PARTITIONS TO ADJACENT MASONRY WALLS.
- DO NOT PLACE OPENINGS ABOVE ANY LINTEL WITHIN A HEIGHT LESS THAN OR EQUAL TO THE WIDTH OF THE CLEAR OPENING BELOW THE LINTEL, UNLESS SPECIFICALLY SHOWN OR APPROVED BY THE SER.
- WHEN INSTALLING POST-INSTALLED ANCHORS (EXPANSION BOLTS, ADHESIVE ANCHORS, ETC.) TAKE MEASURES TO AVOID DAMAGE TO EXISTING REINFORCING STEEL AND MASONRY. CLEAN HOLES IN ACCORDANCE WITH MANUFACTURER'S REQUIREMENTS.

I. STRUCTURAL STEEL

- PREPARE WRITTEN WELD PROCEDURE SPECIFICATIONS (WPS), IN ACCORDANCE WITH AWS D1.1, FOR ALL WELDING PROCEDURES TO BE USED ON PROJECT. A COPY OF ALL WPS SHALL BE MAINTAINED ON SITE AT ALL TIMES.
- OBTAIN CURRENT EVIDENCE OF WELDERS PASSING THE APPROPRIATE AWS QUALIFICATION TESTS. SUCH EVIDENCE MAY BE REQUESTED AT ANY TIME DURING THE PROJECT.
- PROVIDE ACCESS FOR INSPECTION OF ALL SHOP AND FIELD CONNECTIONS FOR PROPER MATERIALS AND WORKMANSHIP.
- USE HIGH STRENGTH BOLTS AND NUTS WITH CLEAR MARKINGS AS REQUIRED BY AISC SPECIFICATIONS. CONNECTIONS MADE WITH UNMARKED BOLTS AND NUTS WILL BE REJECTED.
- TIGHTEN ALL A325 BOLTS TO THE "SMUG TIGHT" CONDITION (DEFINED AS ALL PLIES OF THE CONNECTED MATERIAL HAVE BEEN BROUGHT INTO TIGHT CONTACT).
- WELDS INDICATED AS FIELD WELDS MAY BE MADE IN THE FIELD OR SHOP AT THE CONTRACTOR'S DISCRETION. WELDS INDICATED AS SHOP WELDS SHALL BE MADE IN THE SHOP UNLESS SPECIFICALLY APPROVED OTHERWISE BY THE STRUCTURAL ENGINEER.
- PROVIDE TWO (2) COATS OF BITUMINOUS PAINT ALL STEEL AT AND BELOW FINISHED GRADE OR FLOOR SLAB.
- HOT-DIP GALVANIZE ALL STRUCTURAL STEEL FRAMING WHERE INDICATED IN DRAWINGS.
- NOTIFY THE SER OF ANY FABRICATION OR ERECTION ERRORS OR DEVIATIONS AND RECEIVE WRITTEN APPROVAL BEFORE ANY FIELD CORRECTIONS ARE MADE.
- REPLACE OR REINFORCE ANY STRUCTURAL STEEL DAMAGED DURING CONSTRUCTION (INCLUDING ANCHOR RODS) AS ACCEPTABLE TO THE SER.
- FIELD CUTTING WITH GAS TORCH IS NOT PERMITTED.
- SEE BELOW FOR BAR LAP AND SPLICE SCHEDULE.

BAR SIZE	CONCRETE COMPRESSIVE STRENGTH			
	3000 PSI	4000 PSI	5000 PSI	6000 PSI
#3	17"	15"	13"	12"
#4	22"	19"	17"	16"
#5	28"	24"	22"	20"
#6	33"	29"	26"	24"
#7	48"	42"	38"	34"
#8	55"	48"	43"	39"
#9	62"	54"	48"	44"
#10	70"	61"	54"	50"

- NOTES:
- LAP WELDED WIRE FABRIC THE WIRE SPACING PLUS 6" AT ENDS.
 - ALL LAPS ARE ACI CLASS B UNION.
 - INCREASE SPLICE LENGTHS BY THE FOLLOWING IF:
 - LIGHTWEIGHT CONCRETE IS USED.....+35%
 - HORIZ. TOP BARS W/ GREATER THAN 12" OF CONCRETE BELOW.....+30%
 - BAR SPACING IS LESS THAN 2 BAR DIA.....+50%
 - EPOXY COATED BARS WHEN CLEAR COVER IS LESS THAN 3 BAR DIA. OR CLEAR SPACING IS LESS THAN 6 BAR DIA.....+50%
 - OTHER EPOXY COATED BARS.....+20%

J. STEEL JOISTS AND JOIST GIRDERS

- JOISTS HAVE BEEN SELECTED BASED ON OCCUPANCY LOADS. PERIODICALLY, A SINGLE OCCUPANT, THE LOAD OF WHICH WAS ACCOUNTED FOR IN THE SELECTION OF THE JOISTS, MAY OCCUR AT ANY POINT ALONG THE JOIST LENGTH. THEREFORE, PER THE 2010 SJI CODE OF STANDARD PRACTICE, ALL JOISTS, INCLUDING "SP" JOISTS, SHALL BE DESIGNED FOR ADDITIONAL BENDING STRESSES ("BEND-CHECK") RESULTING FROM A 275 LB. CONCENTRATED LOAD (THE WEIGHT OF AN AVERAGE HUMAN BEING CARRYING CONSTRUCTION MATERIALS) LOCATED AT ANY LOCATION ALONG THE TOP CHORD.
- ANY CONCENTRATED LOAD IN EXCESS OF 100 LB., APPLIED TO THE TOP OR BOTTOM JOIST CHORD AND NOT WITHIN 5' OF THE JOIST BOTTOM CHORD PANEL POINT, REQUIRES INSTALLATION OF TYPICAL JOIST WEB REINFORCEMENT PER THE DETAIL CONTAINED IN THESE DOCUMENTS. PRIOR TO THE APPLICATION OF THE LOAD, NOTIFY THE SER IF THE LOAD WILL EXCEED 350 LB. FOR K-SERIES JOISTS OR 600 LB. FOR LH- OR DLH-SERIES JOISTS.
- SPECIAL JOISTS (SP), WHERE INDICATED ON THE PLANS, HAVE SPECIAL DESIGN REQUIREMENTS. REFER TO PLANS AND DETAILS FOR LOCATIONS AND LOADING DIAGRAMS.
- TEMPORARILY SHORE JOIST (DESIGN BY SPECIALTY CONTRACTOR) WHEN INSTALLING FILLET WELDS TO IT, NEAR THE POINT OF WELDING.
- WELD OR BOLT BRIDGING AND ANCHOR AT END WALLS OR BEAMS, IN ACCORDANCE WITH THE APPROVED SHOP DRAWINGS AND SJI SPECIFICATIONS.
- JOIST MANUFACTURER SHALL DESIGN AND DETAIL BRIDGING FOR ERECTION AND PERMANENT CONDITIONS. THE BRIDGING SYSTEM SHALL BE DESIGNED AND DETAILED TO BE INTERNALLY STABLE WITHIN THE JOIST SYSTEM WITHOUT RELIANCE ON MASONRY WALLS, STEEL BEAMS OR OTHER STRUCTURAL ELEMENTS FOR STABILITY.
- INSTALL AND PERMANENTLY FASTEN BRIDGING, BRIDGING ANCHORS AND JOIST CONNECTIONS COMPLETELY PRIOR TO THE APPLICATION OF ANY CONSTRUCTION LOADS.
- FOR ROOF JOISTS RESISTING NET WIND UPLIFT, PROVIDE BRIDGING OF BOTTOM CHORD AT THE FIRST PANEL POINT FROM SUPPORTS. PROVIDE ADDITIONAL BRIDGING AS REQUIRED BY THE JOIST MANUFACTURER.
- SUPPORT ROOFTOP UNITS AND OTHER SUSPENDED EQUIPMENT AND PIPING DIRECTLY FROM JOIST PANEL POINTS OR PROVIDE ADDITIONAL WEB REINFORCEMENT AS SHOWN IN THE DRAWINGS.

K. STEEL DECK

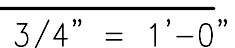
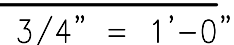
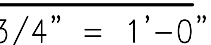
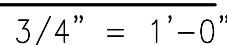
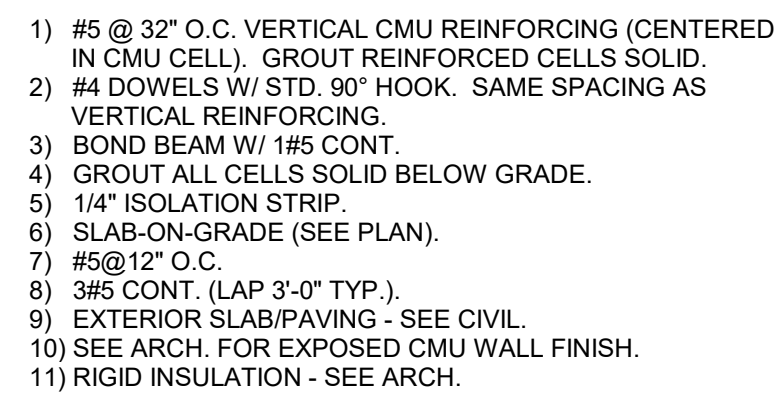
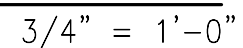
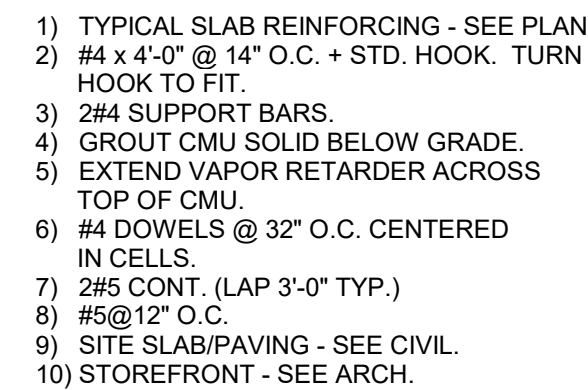
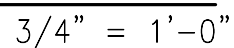
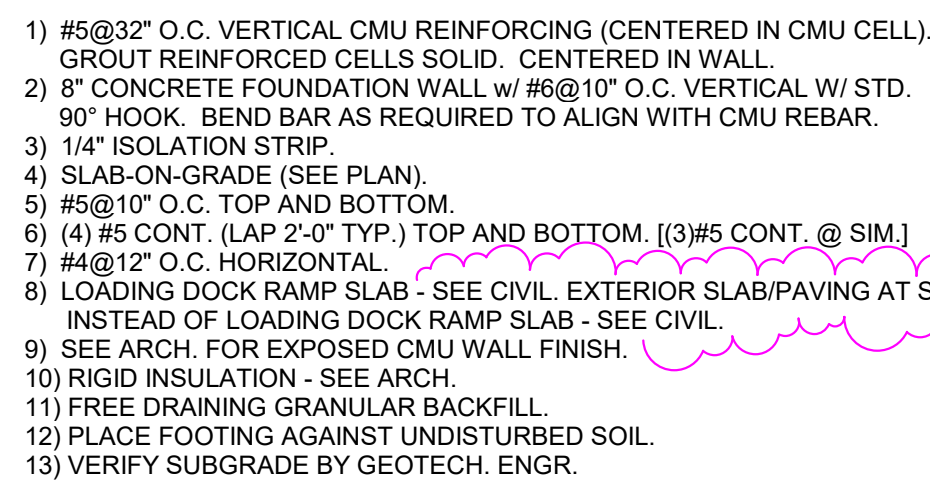
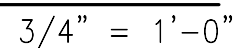
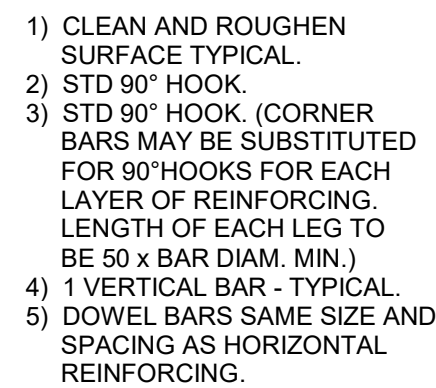
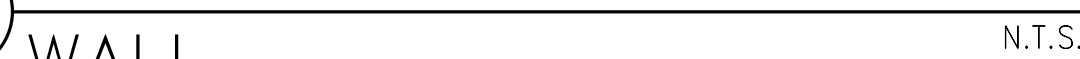
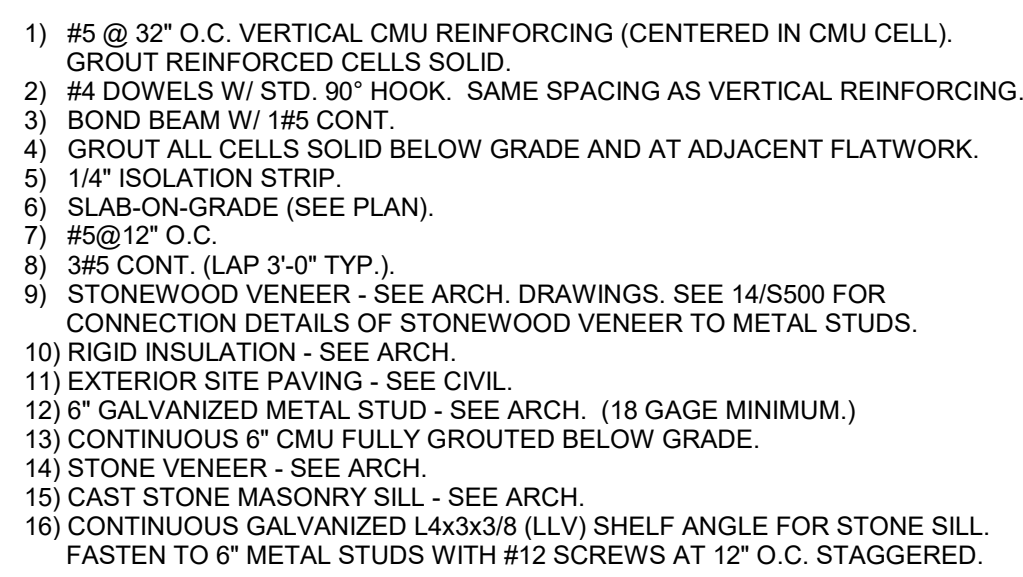
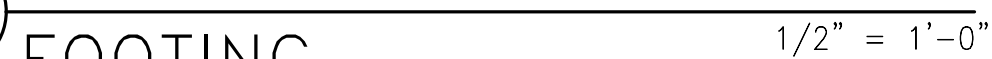
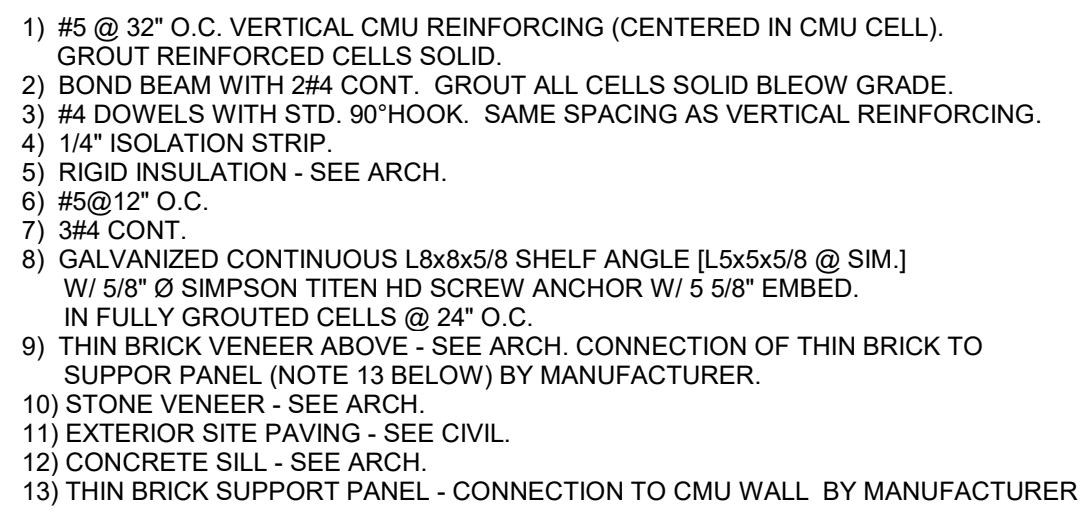
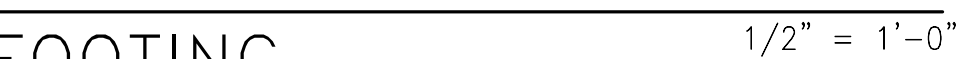
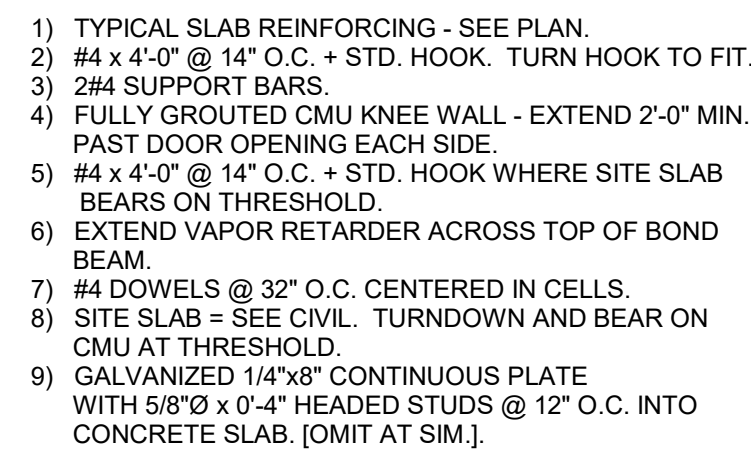
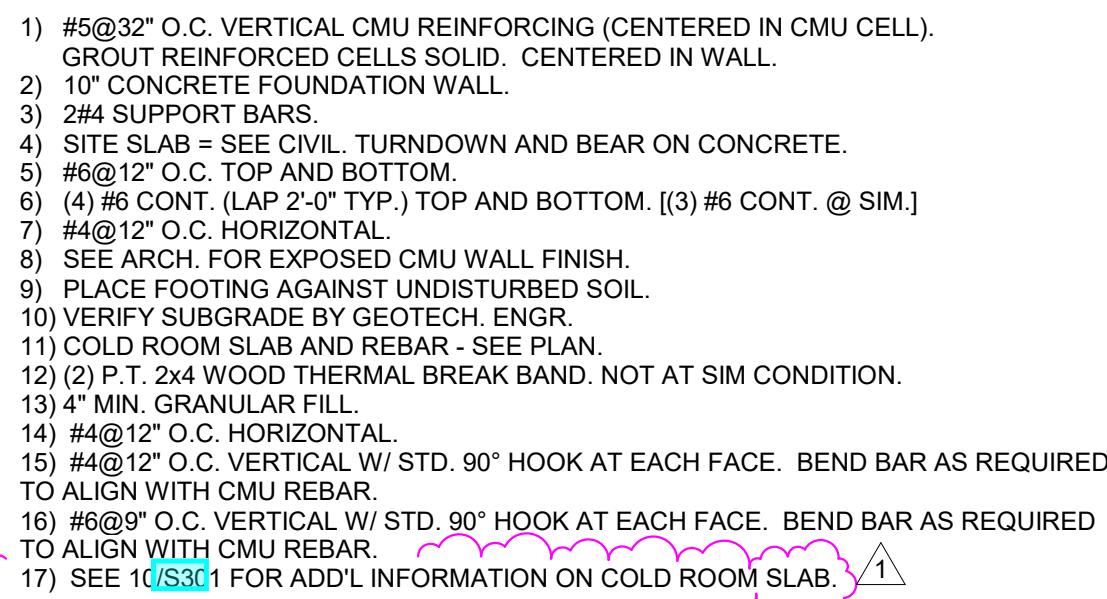
- PROVIDE DECKS MEETING THE CRITERIA INDICATED IN THESE NOTES AND THE PROJECT SPECIFICATIONS. SUBMIT MANUFACTURER'S LOAD TABLES AND PROPERTY DATA FOR REVIEW AND APPROVAL BY THE SER.
- BASIS OF DESIGN DECKING PROPERTIES ('I' AND 'S')



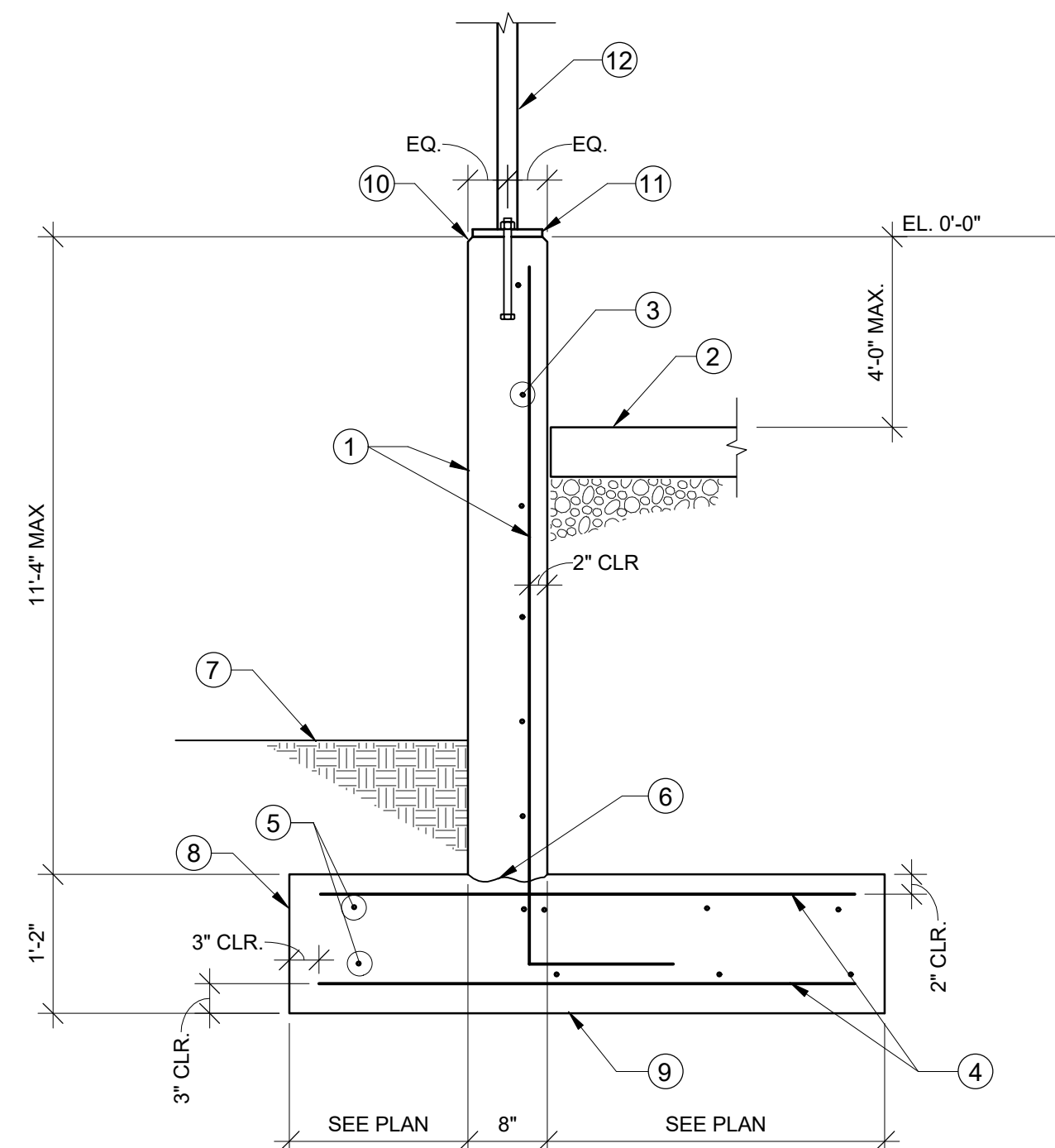
- ## STRUCTURAL ELEVATIONS S200



ABBREVIATIONS: EWB = EACH WAY BOTTOM

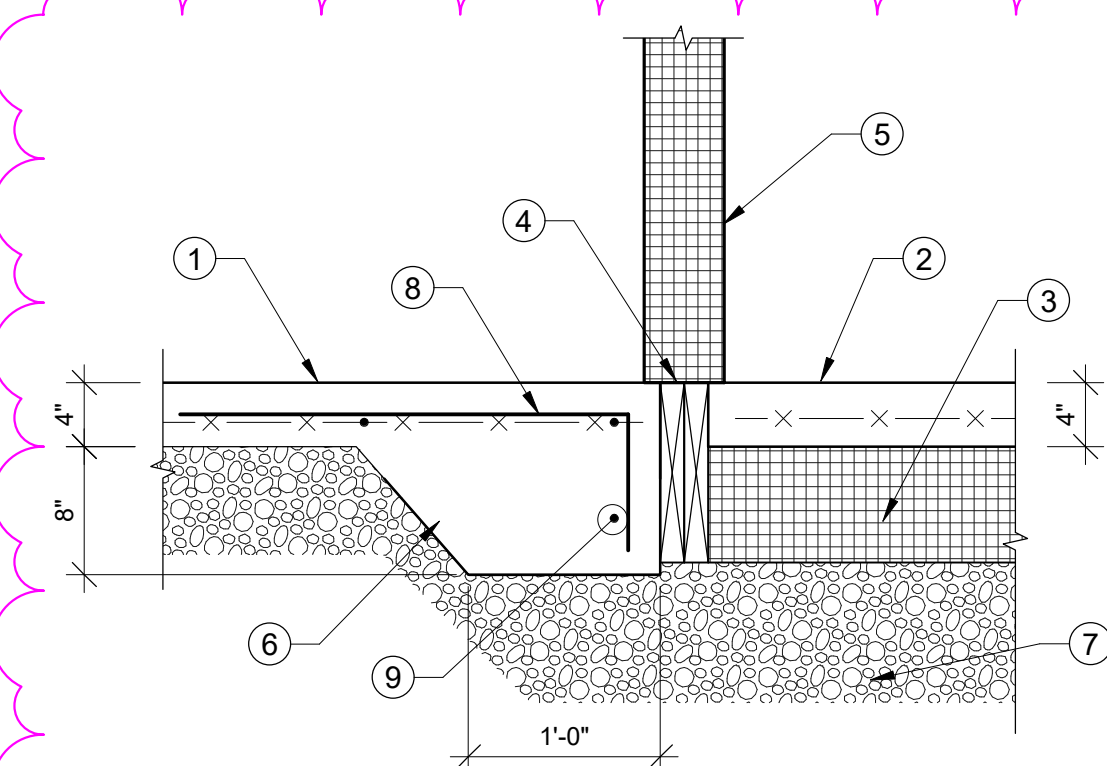


7 LOADING DOCK RETAINING WALL
3/4" = 1'-0"



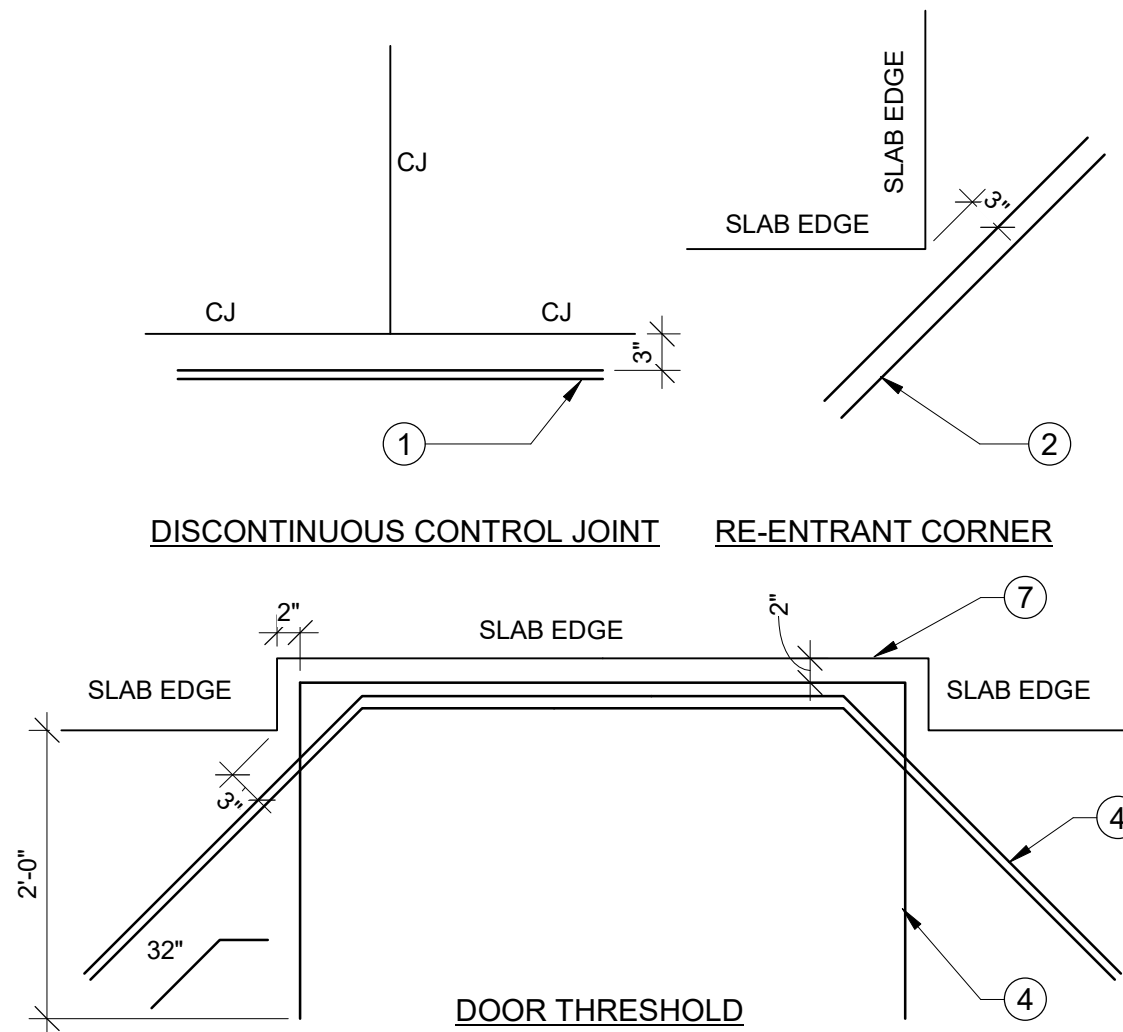
- 1) 8" CONCRETE FOUNDATION WALL W/ #5@12" O.C. INSIDE FACE VERTICAL.
- 2) LOADING DOCK RAMP SLAB - SEE CIVIL.
- 3) #4@12" O.C. HORIZONTAL INSIDE FACE.
- 4) #5@12" O.C. TOP AND BOTTOM.
- 5) (5) #5 CONTINUOUS TOP AND BOTTOM. ((3) #5 CONT. @ SIM.)
- 6) ROUGHEN SURFACE.
- 7) FREE DRAINING GRANULAR BACKFILL.
- 8) PLACE FOOTING AGAINST UNDISTURBED SOIL.
- 9) VERIFY SUBGRADE BY GEOTECH. ENGR.
- 10) CHAMFER CORNERS TYP.
- 11) GALVANIZED 5/8"x6"x6" PLATE WITH GALVANIZED (2)3/4"x9" ANCHOR RODS WITH 6" MIN. EMBED. AT EACH HANDRAIL POST. SEE ARCH. FOR POST LOCATIONS.
- 12) GALVANIZED HANDRAIL POSTS - SEE ARCH. (MAX. 4'-0" O.C.).

3 FREEZER ROOM SLAB
1" = 1'-0"



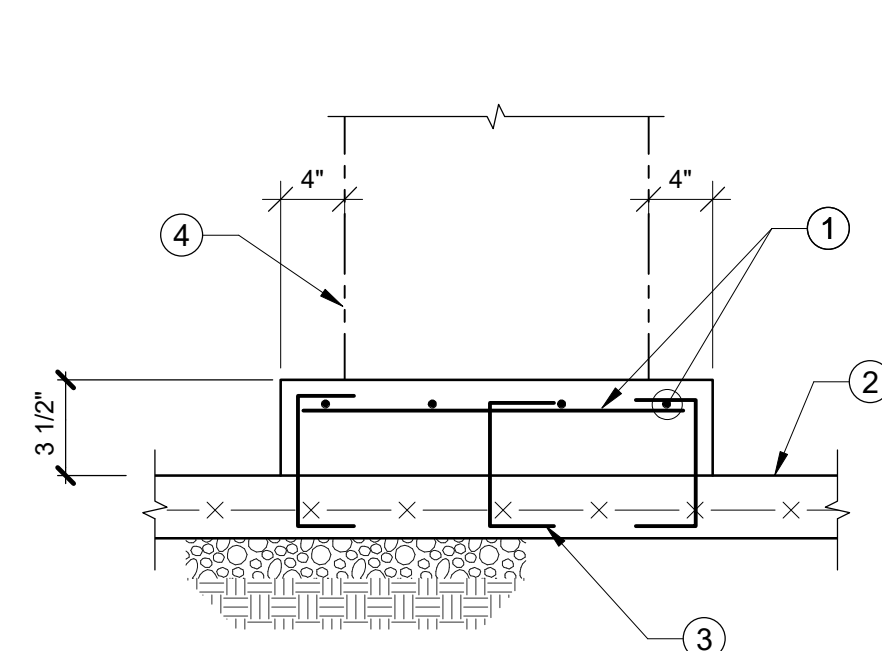
- 1) SLAB TURNDOWN AT COOLER/FREEZER.
- 2) COOLER/FREEZER ROOM SLAB AND REBAR - SEE PLAN.
- 3) RIGID INSULATION. 25 PSI RATING - SEE ARCH.
- 4) (2) P.T. 2x12 WOOD THERMAL BREAK BAND.
- 5) COOLER/FREEZER WALL - SEE ARCH.
- 6) TYPICAL SLAB VAPOR BARRIER.
- 7) 4" MIN. GRANULAR FILL.
- 8) #4 HOOKED DOVEL @ 14" O.C. LAP WITH SLAB REBAR 2'-1" MIN.
- 9) PROVIDE (1)#4 BAR CONTINUOUS.

2 TYP. SLAB-ON-GRADE REINF.
N.T.S.



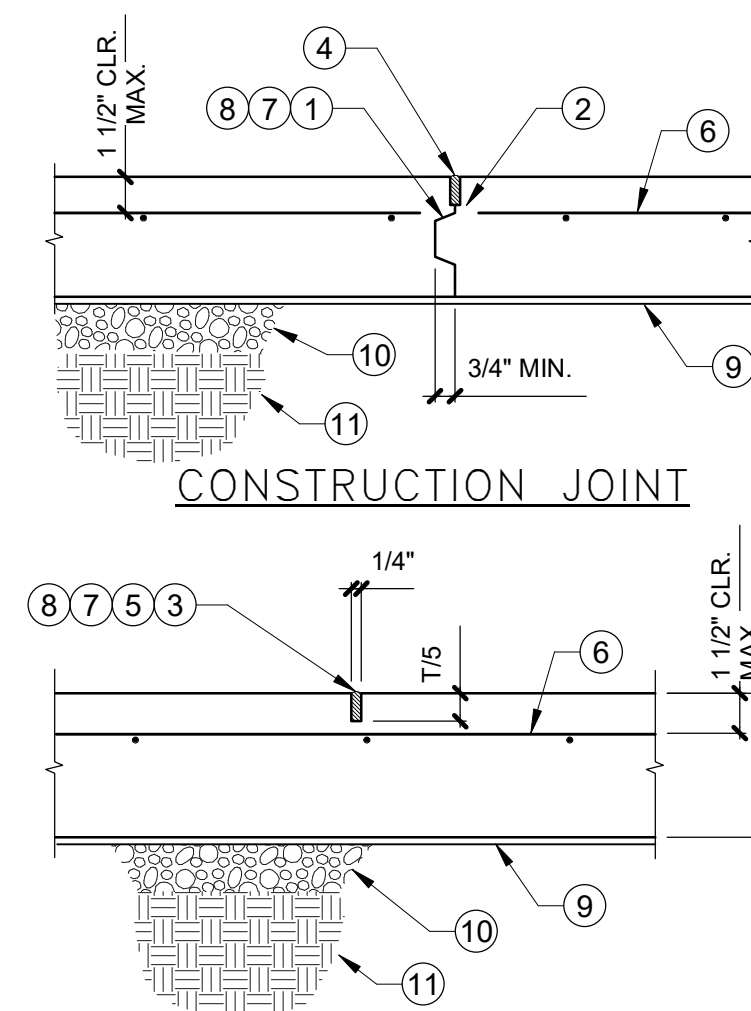
- 1) (2) - #5 x 4'-0" LONG.
- 2) (2) - #5 x 4'-0" LONG.
- 3) (2)#5 2'-0" BARS AT EACH CORNER OF COLUMN.
- 4) (2) #5
- 5) (2)#5 2'-0" BARS AT EACH CORNER OF COLUMN.
- 6) 1/4" ISOLATION JOINT AROUND COLUMN.
- 7) ARMOR-EDGE STEEL JOINT - SEE ARCH.

5 CONCRETE PAD FOR INTERIOR EQUIPMENT
N.T.S.



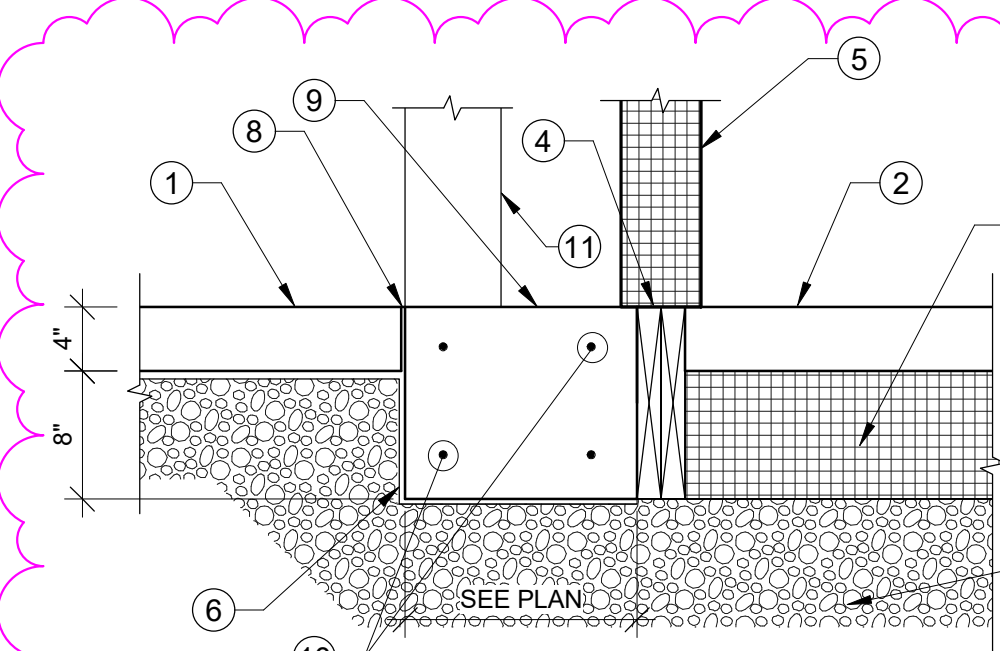
- 1) #4@12" EACH WAY.
- 2) SLAB ON GRADE - SEE PLAN.
- 3) #4 DOWELS @ 24" EACH WAY (MIN. OF ONE EACH CORNER).
- 4) ELECTRICAL EQUIPMENT - VERIFY ALL PAD LOCATIONS AND DIMENSIONS WITH THE M/E/P ENGINEER AND REQUIREMENTS OF THE ACTUAL EQUIPMENT PURCHASED.

1 SLAB ON GRADE JOINTS
N.T.S.



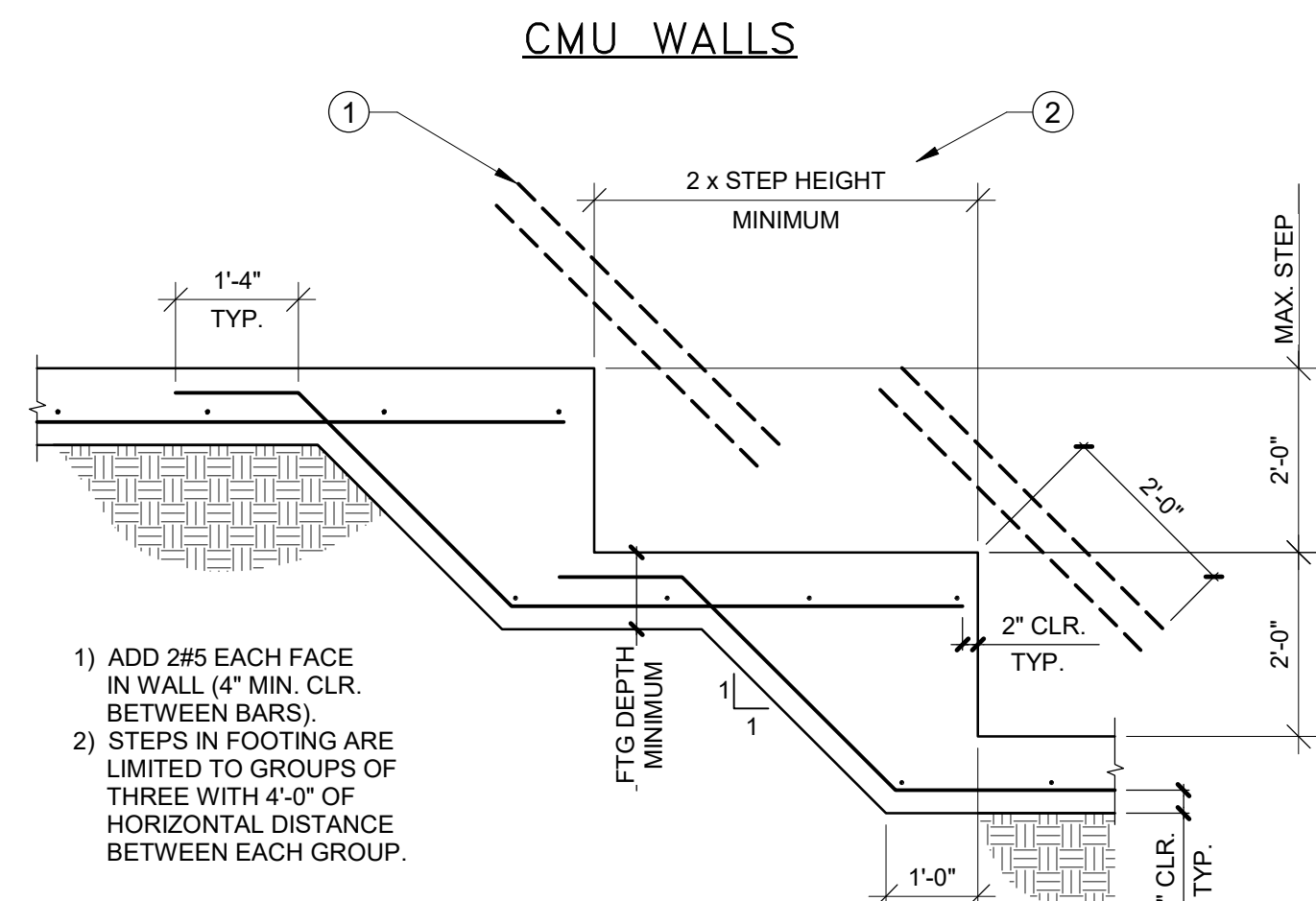
- 1) WOOD FRAMED OR PROPRIETARY PRE-FABRICATED EDGE FORM.
- 2) NO REINFORCING THRU JOINT AT CONSTRUCTION JOINTS.
- 3) TOOLED OR SAWCUT JOINTS ARE PERMITTED. AT THE CONTRACTOR'S OPTION. IF USED, SAWCUT JOINTS USING A "SOFF-CUT" MACHINE OR EQUAL IMMEDIATELY AFTER FINISHING SLAB.
- 4) TOOL JOINT. PLACE BOND BREAKER AND FILL WITH CLASS 50 SILICONE SEALANT. DELAY PLACEMENT OF SEALANT 14 DAYS AFTER CONCRETE PLACEMENT.
- 5) PLACE BACKER ROD AND FILL JOINT WITH CLASS 50 SILICONE SEALANT. DELAY PLACEMENT OF SEALANT A MINIMUM OF 14 DAYS AFTER CONCRETE PLACEMENT.
- 6) REBAR - PROVIDE SAND PLATE SUPPORT CHAIRS TO HOLD REINFORCING IN POSITION DURING CONCRETE PLACEMENT.
- 7) SEE SLOTT FOR CONTROL JOINT LOCATIONS AND SEE BLUE FOR SLAB POUR SEQUENCE.
- 8) PROVIDE CONSTRUCTION JOINT ONLY WHERE NOTED ON PLAN.
- 9) VAPOR RETARDER.
- 10) GRANULAR FILL PER PROJECT GEOTECHNICAL REPORT.
- 11) SUBGRADE.

10 FREEZER ROOM SLAB
1" = 1'-0"



- 1) TYPICAL SLAB AND REINFORCING - SEE PLAN.
- 2) COOLER/FREEZER ROOM SLAB AND REBAR - SEE PLAN.
- 3) RIGID INSULATION. 25 PSI RATING - SEE ARCH.
- 4) (2) P.T. 2x12 WOOD THERMAL BREAK BAND.
- 5) COOLER/FREEZER WALL - SEE ARCH.
- 6) TYPICAL SLAB VAPOR BARRIER.
- 7) 4" MIN. GRANULAR FILL.
- 8) CONSTRUCTION JOINT - SEE PLAN.
- 9) THICKENED SLAB BETWEEN COOLER/FREEZER AND CONSTRUCTION JOINT. PROVIDE CONTROL JOINTS AT 4'-0" O.C.
- 10) PROVIDE (2)#4 BAR CONTINUOUS TOP AND BOTTOM.
- 11) INTERIOR PARTITION - SEE ARCH.

9 STEPPED FOOTING
1/2" = 1'-0"

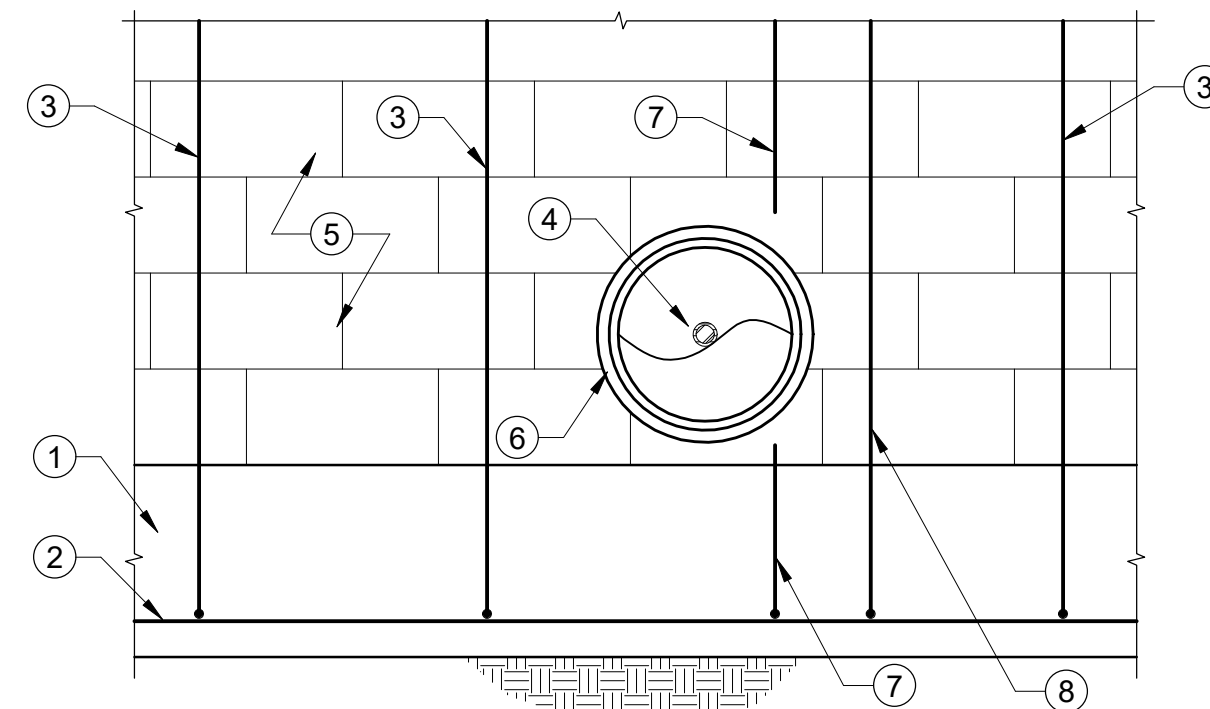


- 1) ADD 2#5 EACH FACE IN WALL (4" MIN. CLR. BETWEEN BARS).
- 2) STEPS IN FOOTING ARE LIMITED TO GROUPS OF THREE WITH 4'-0" OF HORIZONTAL DISTANCE BETWEEN EACH GROUP.

CONCRETE WALLS

CMU WALLS

8 UTILITY PENETRATION THROUGH CMU WALL
N.T.S.



- 1) WALL FOOTING.
- 2) WALL FOOTING REINFORCING STEEL.
- 3) VERTICAL DOWELS AND REINFORCEMENT - SEE WALL DETAILS.
- 4) UTILITY PIPING.
- 5) FULLY GROUTED CMU FOUNDATION WALL.
- 6) SLEEVED PENETRATION WITH MIN. 1" COMPRESSIBLE FILL. SEE ARCH. FOR WATERPROOFING.
- 7) INTERRUPTED VERTICAL REINFORCEMENT AT PENETRATION (WHERE OCCURS).
- 8) OFFSET AND ADD FULL HEIGHT MATCHING VERT. BAR(S) TO REPLACE INTERRUPTED BAR(S).

